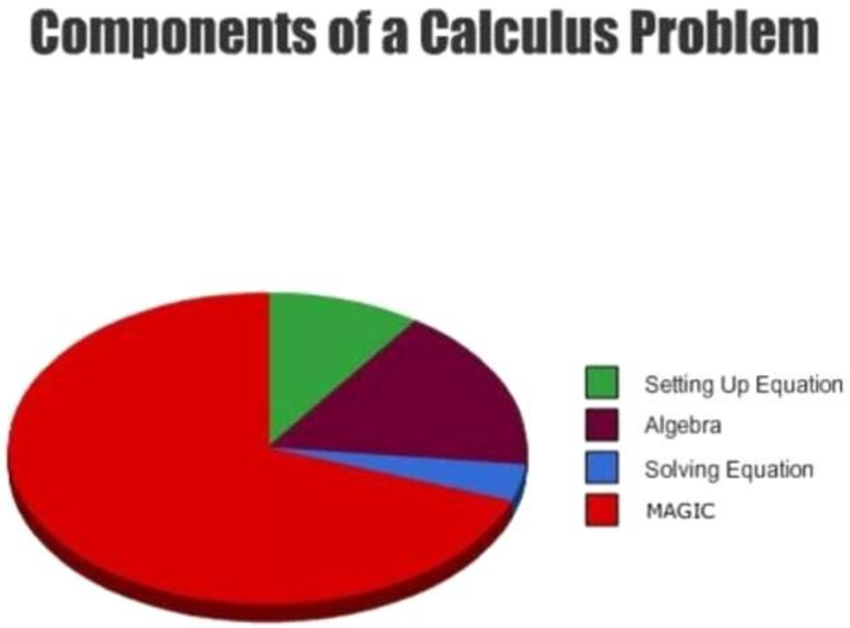
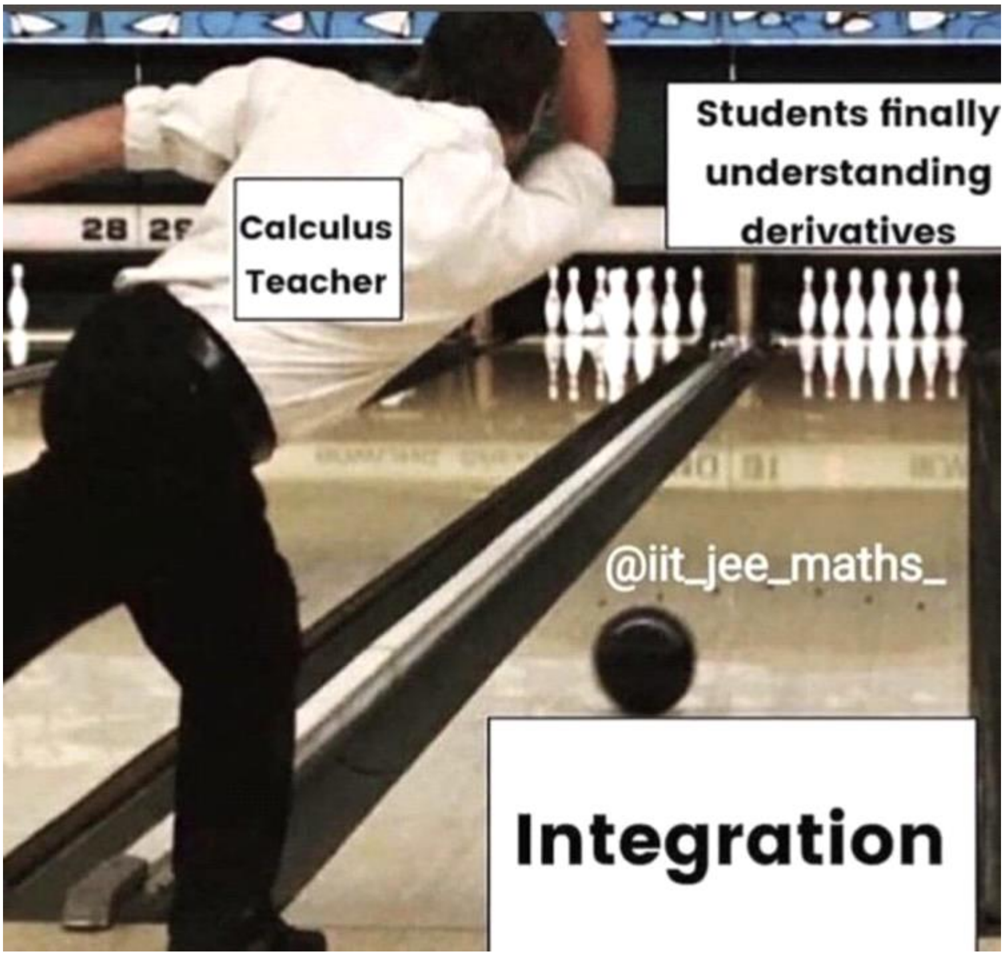
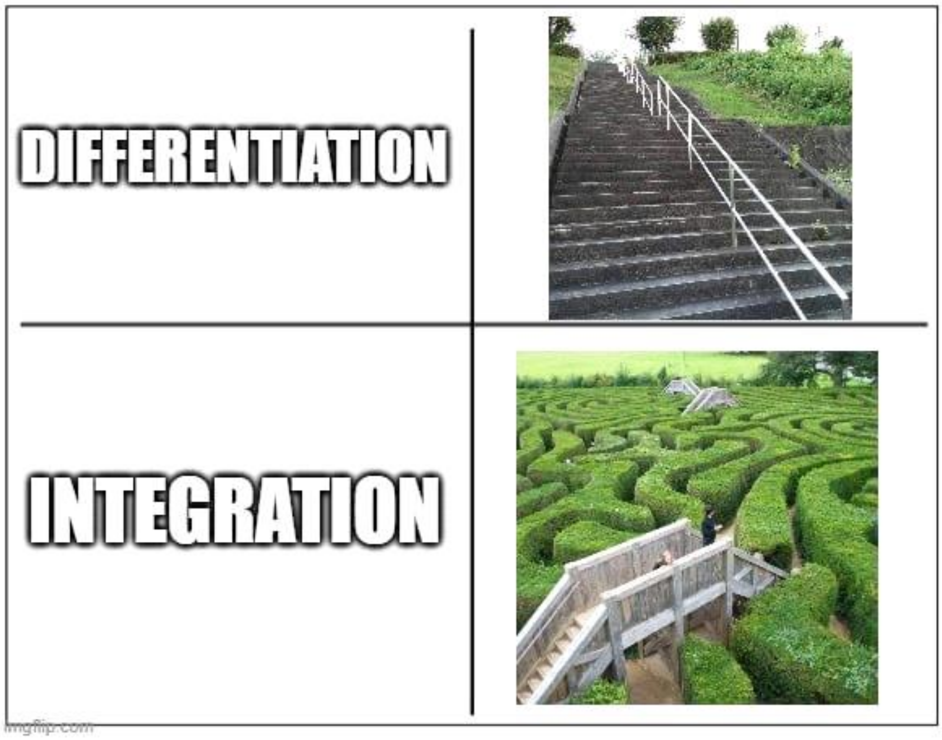
Lec-06-Week-04

Saturday, 22 August 2026 11:44 am






DIFFERENTIATION
INTEGRATION
imgflip.com

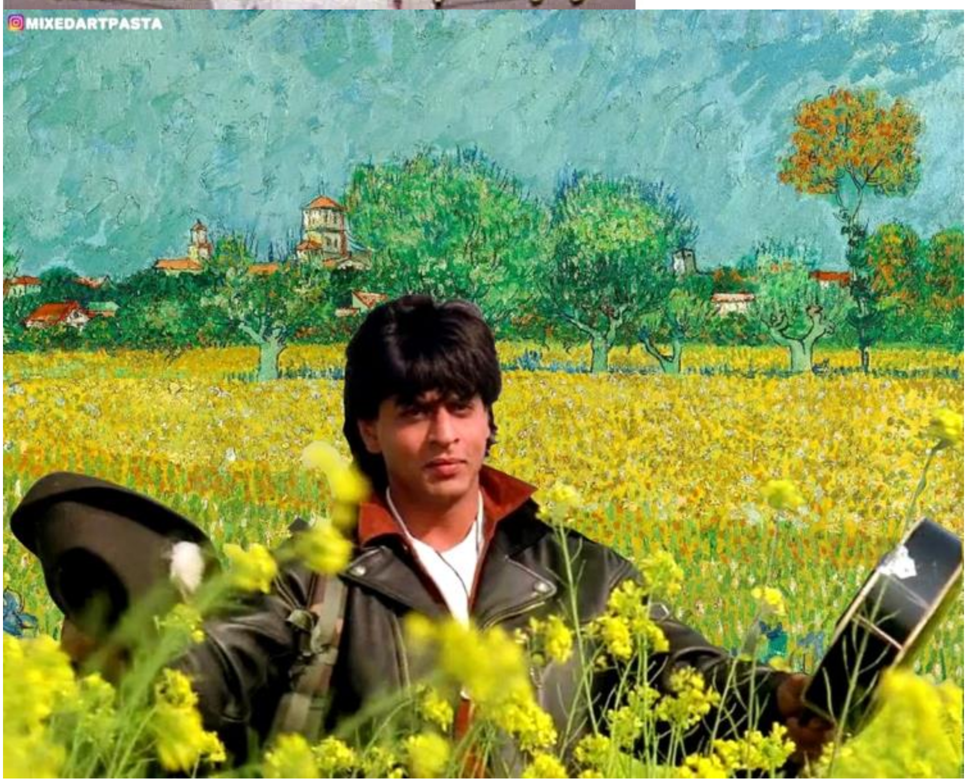
Differentiation

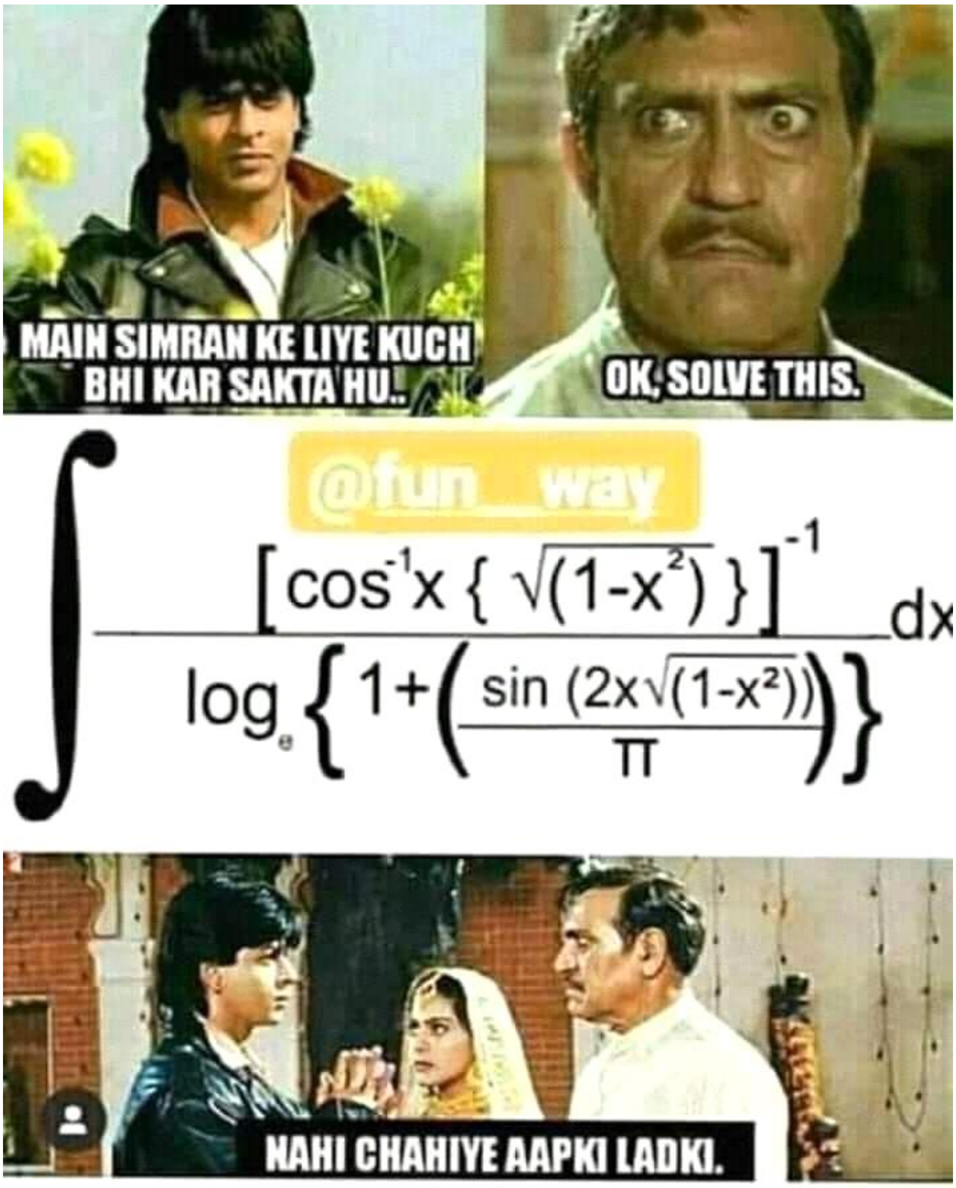
Simple integration

Integration by substitution

Integration by parts

Literally every type of integration in existence

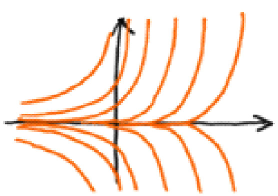




GENERAL SOLUTION

Diff. Eq. : $\frac{dy}{dt} = 2y$

General Solution : $y = Ce^{2t}$

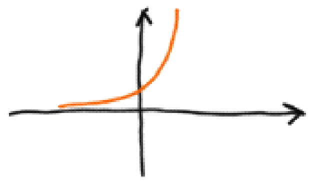


Includes all possible solutions (and no functions that are not solutions)

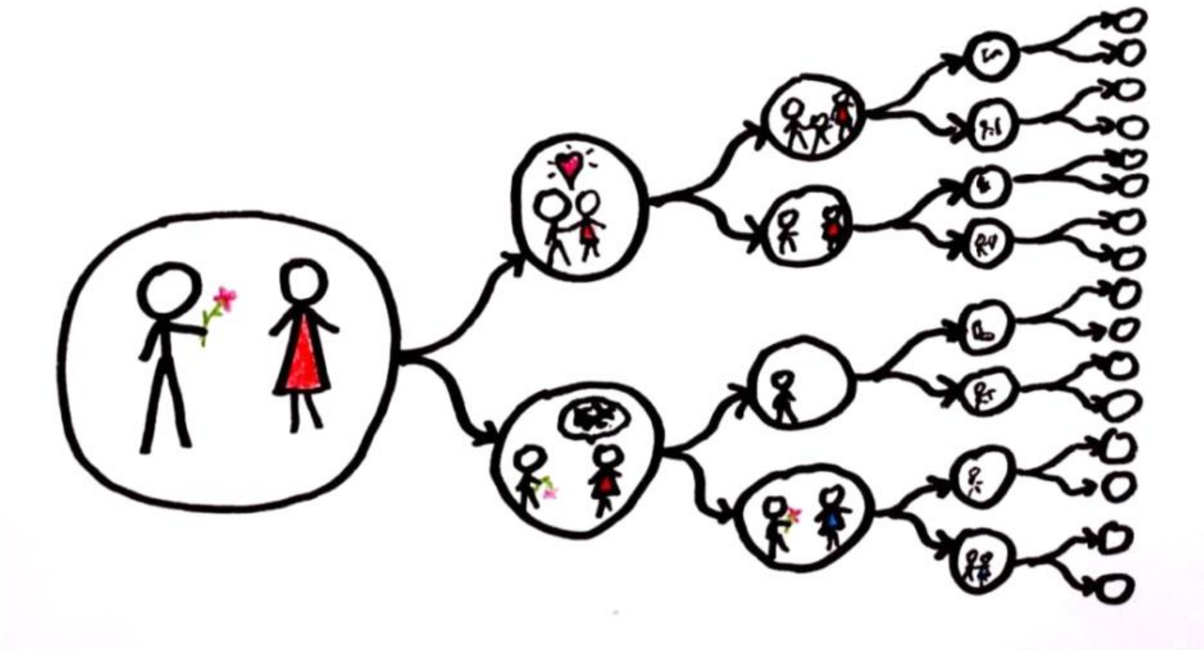
PARTICULAR SOLUTION

Diff. Eq. : $\frac{dy}{dt} = 2y$

Particular Solution : $y = e^{2t}$ $(C=1)$



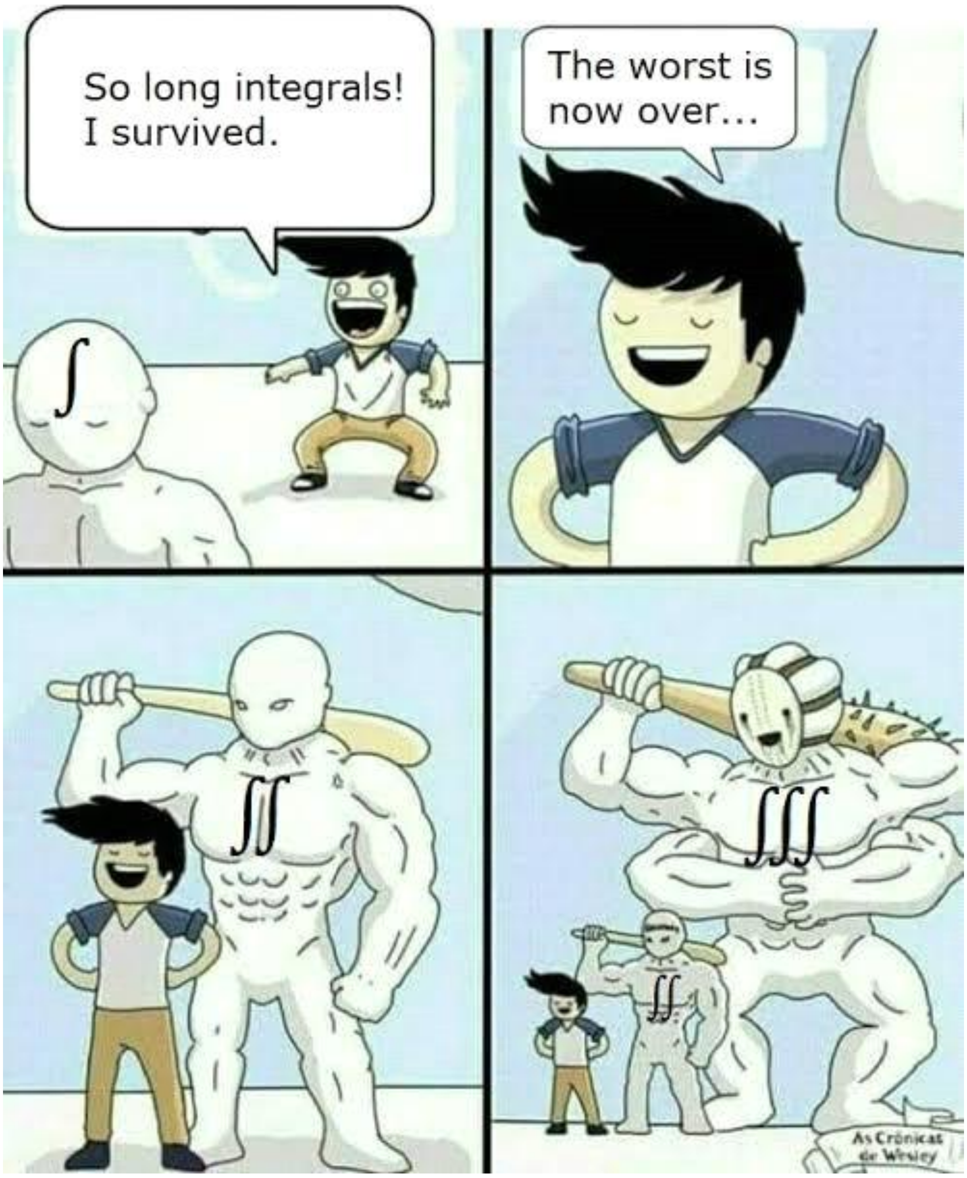
One individual solution whose graph is a single curve (an integral curve)



$$\int_a^b \int_c^d f(x,y)\,dx\,dy \quad \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_i,y_j)\Delta x \Delta y$$



+
Σ
∫
∭
Upgrade
Upgrade
go back
I SAID GO BACK


So long integrals! I survived.
The worst is now over...
∫
∬
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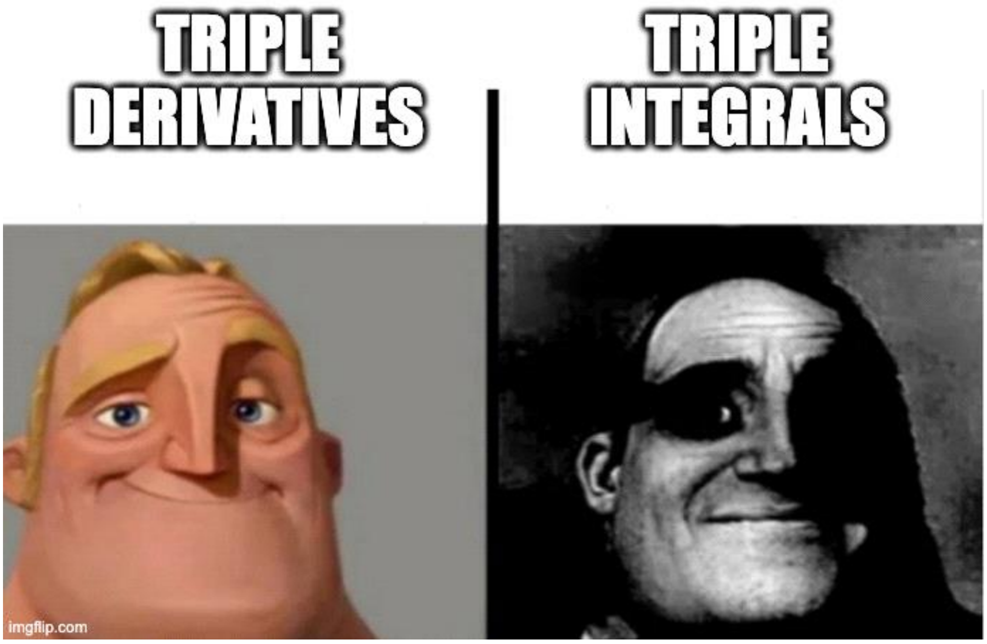
∬
As Crônicas de Wesley


me
double integrals
triple integrals


TRIPLE DERIVATIVES
TRIPLE INTEGRALS
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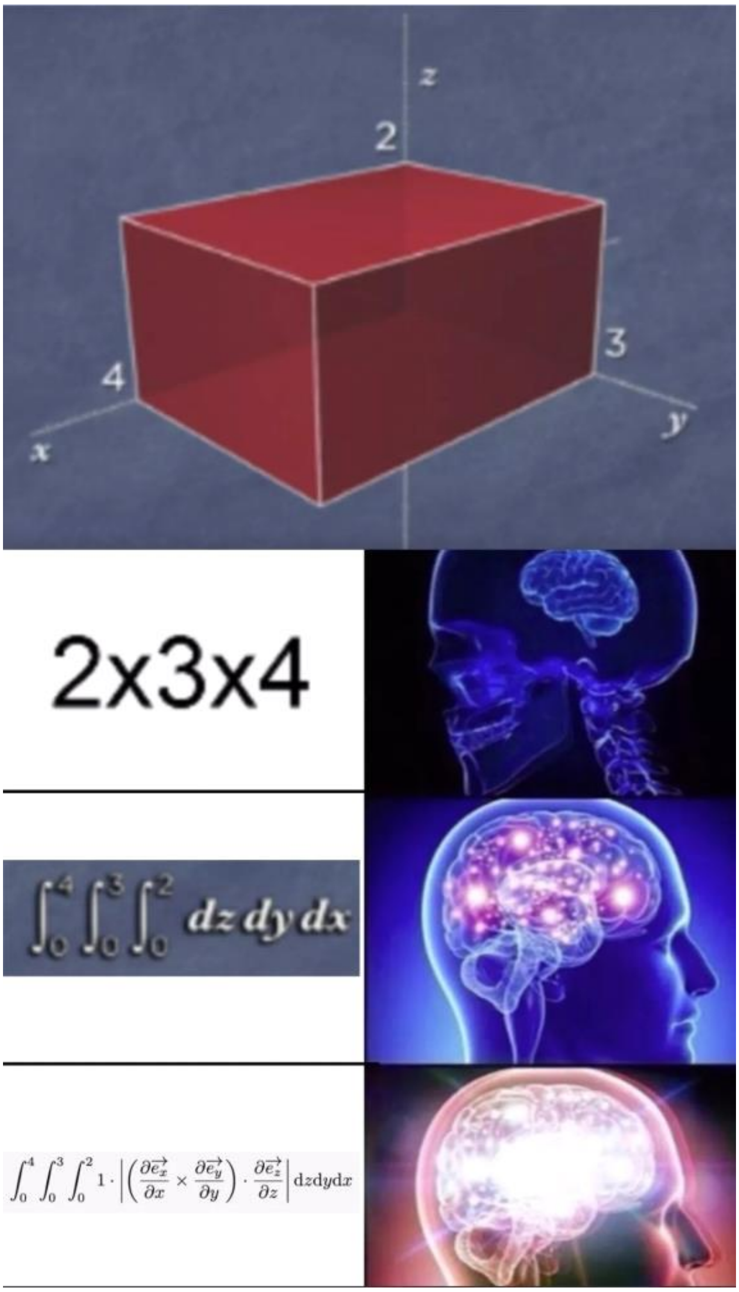


sketch the region of integration.

1. $\displaystyle\int_0^{\pi}\int_0^{x} y\sin x\,dy\,dx$ **2.** $\displaystyle\int_0^{1}\int_{y^2}^{y} xy\,dx\,dy$

evaluate the integral.

9. $\displaystyle\int_0^{1}\int_0^{1} ye^{xy}\,dx\,dy$

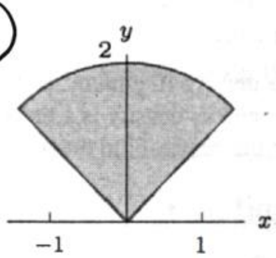
37. $\displaystyle\int_0^1 \int_{e^y}^{e} \frac{x}{\ln x}\, dx\, dy$

27. $\int_R (2x+3y)^2\, dA$, where R is the triangle with vertices at $(-1,0)$, $(0,1)$, and $(1,0)$.

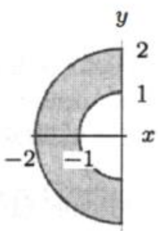
39. (a) Find the volume below the surface $z = x^2 + y^2$ and above the xy-plane for $-1 \le x \le 1, -1 \le y \le 1$.
(b) Find the volume above the surface $z = x^2 + y^2$ and below the plane $z = 2$ for $-1 \le x \le 1$, $-1 \le y \le 1$.

For the regions R in Exercises 1–4, write $\int_R f\, dA$ as an iterated integral in polar coordinates.

3.



4.



Sketch the region of integration in Exercises

12. $\displaystyle\int_{\pi/6}^{\pi/3} \int_0^1 f(r,\theta)\, r\, dr\, d\theta$

15. $\displaystyle\int_{\pi/4}^{\pi/2} \int_0^{2/\sin\theta} f(r,\theta)\, r\, dr\, d\theta$

16. $\int_R \sqrt{x^2+y^2}\, dx dy$ where R is $4 \le x^2 + y^2 \le 9$.

Convert the integrals in Problems 19–21 to polar coordinates and evaluate.

19. $\displaystyle\int_{-1}^{0}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} x\,dy\,dx$

21. $\displaystyle\int_{0}^{\sqrt{2}}\int_{y}^{\sqrt{4-y^2}} xy\,dx\,dy$

24. Evaluate the integral by converting it into Cartesian coordinates:
$$\int_0^{\pi/6}\int_0^{2/\cos\theta} r\,dr\,d\theta.$$

25. (a) Sketch the region of integration of
$$\int_0^1\int_{\sqrt{1-x^2}}^{\sqrt{4-x^2}} x\,dy\,dx+\int_1^2\int_0^{\sqrt{4-x^2}} x\,dy\,dx$$

(b) Evaluate the quantity in part (a).

26. Find the volume of the region between the graph of $f(x,y)=25-x^2-y^2$ and the xy plane.

Sketch the region of integration in Exercises 5–13.

5. $\displaystyle\int_0^1\int_{-1}^1\int_0^{\sqrt{1-x^2}} f(x,y,z)\,dz\,dx\,dy$

8. $\displaystyle\int_{-1}^1\int_0^1\int_{-\sqrt{1-z^2}}^{\sqrt{1-z^2}} f(x,y,z)\,dy\,dz\,dx$

9. $\displaystyle\int_{-1}^1\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\int_0^{\sqrt{1-x^2-z^2}} f(x,y,z)\,dy\,dz\,dx$

13. $\displaystyle\int_0^1\int_0^{\sqrt{1-z^2}}\int_{-\sqrt{1-x^2-z^2}}^{\sqrt{1-x^2-z^2}} f(x,y,z)\,dy\,dx\,dz$

31. A trough with triangular cross-section lies along the x-axis for $0 \leq x \leq 10$. The slanted sides are given by $z = y$ and $z = -y$ for $0 \leq z \leq 1$ and the ends by $x = 0$ and $x = 10$, where x, y, z are in meters. The trough contains a sludge whose density at the point (x, y, z) is $\delta = e^{-3x}$ kg per m^3.

(a) Express the total mass of sludge in the trough in terms of triple integrals.

(b) Find the mass.

In Exercises 2–7, find an equation for the surface.

2. The vertical plane $y = x$ in cylindrical coordinates.

3. The top half of the sphere $x^2 + y^2 + z^2 = 1$ in cylindrical coordinates.

4. The cone $z = \sqrt{x^2 + y^2}$ in cylindrical coordinates.

5. The cone $z = \sqrt{x^2 + y^2}$ in spherical coordinates.

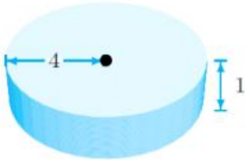
8. $f(x, y, z) = \sin(x^2 + y^2)$, W is the solid cylinder with height 4 and with base of radius 1 centered on the z axis at $z = -1$.

In Exercises 10–11, evaluate the triple integrals in spherical coordinates.

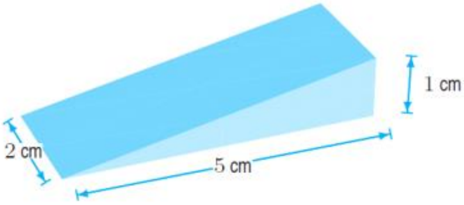
10. $f(\rho, \theta, \phi) = \sin\phi$, over the region $0 \leq \theta \leq 2\pi$, $0 \leq \phi \leq \pi/4$, $1 \leq \rho \leq 2$.

11. $f(x, y, z) = 1/(x^2 + y^2 + z^2)^{1/2}$ over the bottom half of the sphere of radius 5 centered at the origin.

For Exercises 12–18, choose coordinates and set up a triple integral, including limits of integration, for a density function f over the region.



17.



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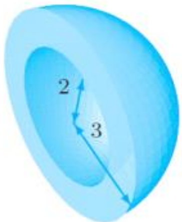
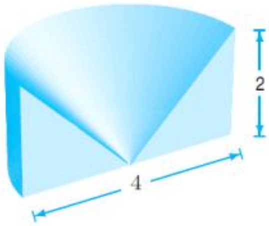


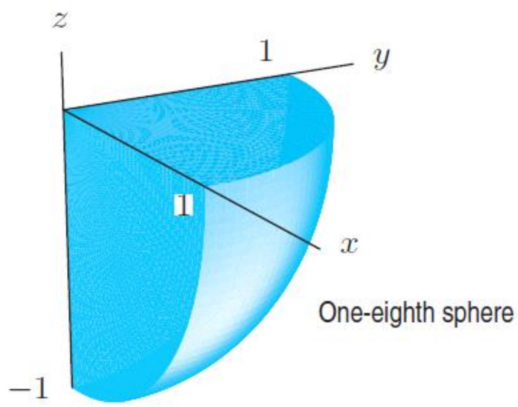
18.



For the regions W shown in Problems 24–26, write the limits of integration for $\int_W dV$ in the following coordinates:

(a) Cartesian **(b)** Cylindrical **(c)** Spherical

24.



39. Suppose W is the region outside the cylinder $x^2+y^2 = 1$ and inside the sphere $x^2 + y^2 + z^2 = 2$. Calculate

$$\int_W (x^2 + y^2)\, dV.$$

32. Between the cone $z = \sqrt{x^2 + y^2}$ and the first quadrant of the xy-plane, with $x^2 + y^2 \leq 7$.